Duration: 3hrs **[Max Marks:80]**

N.B. : (1) Question No 1 is Compulsory.
(2) Attempt any three questions out of the remaining five.
(3) All questions carry equal marks.
(4) Assume suitable data, if required and state it clearly.

1 Attempt any FOUR **[20]**

a Define each software testing terminology:
i) Failure, ii) Defect, iii)Error, iv)Testware and v)Test oracle.

b What is Mutation testing? Differentiate between primary and secondary mutants.

c What criteria you will consider for selection of test tools for automation Testing.

d Explain structure of testing Group.

e Discuss Six Sigma.

2 a Consider a project with the following distribution of data and calculate its defect spoilage. **[10]**

SDLC Phase	No. of Defects	Defect Age
Requirement Specs.	34	2
HLD	25	3
LLD	17	7
Coding	10	8

b Explain Agile Testing Life Cycle and its challenges. **[10]**

3 a A program reads three numbers A, B and C, within the range [1,100] and prints the minimum number. Design test cases for this program using BVC and Robust testing methods. **[10]**

b What is the need of software measurement? Discuss the various types of software metrics. **[10]**

4 a What is the need of automation testing activities? Differentiate between static and dynamic tools? **[10]**

b Consider following C code. **[10]**

```
main()
{
    int number, index;
    1. printf("Enter a number");
    2. scanf("%d",&number);
    3. index=2;
    4. while(index<=number-1)
```


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```
5.  {
6.      if(number%index==0)
7.      {
8.          printf("Not a prime number");
9.          break;
10.     }
11.    index++;
12. }
13. if(index==number)
14.    printf("prime number");
15. } // end main
```

Draw DD graph, Calculate cyclomatic complexity, List out independent paths and design test cases.

5 a What are the components of a test plan? Illustrate test plan hierarchy with a neat diagram. **[10]**

b Explain McCall's Quality factors and Criteria. **[10]**

6 a Explain a bug life cycle with a neat diagram in detail. List down the states of a bug. **[10]**

B Differentiate between Effective Software Testing and Exhaustive Software Testing. **[10]**

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1 Attempt any FOUR **[20]**

a Explain Bottom up Integration Testing with one example

b How to test simple loop in a program?

c Explain Logic Coverage Criteria for Statement Coverage.

d Explain State Table Based testing with example.

e Discuss the features and use of Bugzilla Testing Tool.

2 a Consider the program for calculating the factorial of a number. **[10]**

```
main()
{
int number;
int fact();
1. clrscr();
2. printf("Enter the number whose factorial is to be found out");
3. scanf("%d", &number);
4. if(number <0)
5. printf("Facorial cannot be defi ned for this number);
6. else
7. printf("Factorial is %d", fact(number));
8. }
int fact(int number)
{
int index;
1. int product =1;
2. for(index=1; index<=number; index++)
3. product = product * index;
4. return(product);
5. }
```

(A) Draw the DD graph for the program.
(B) Calculate the individual cyclomatic complexity number for main() and fact() and then, the cyclomatic complexity for the whole program.

b What is Agile testing? Explain challenges in Agile Testing **[10]**

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3	a	Explain the process of test suite prioritization with its type in detail with example.	**[10]**
	b	What are the Key elements of Test Management? Explain the structure of testing group.	**[10]**
4	a	What are graph metrics? Explain with an example how to calculate cyclomatic complexity using graph metrics.	**[10]**
	b	Why is automation important in testing? What are the guidelines for selecting testing tools?	**[10]**
5	a	What is Mutation Testing? Explain primary mutant with an example	**[10]**
	b	Classify different types of bugs based on the Software development lifecycle with example	**[10]**
6	a	Explain ISO 9000:2000 standard.	**[10]**
	b	What is performance testing? Why is performance testing important?	**[10]**

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